

Haley N. Scolati

University of British Columbia | Dept. of Chemistry | hnscolati@chem.ubc.ca

linkedin.com/in/haley-scolati-936458300/ | ORCID: 0000-0002-8505-4459 | github.com/HumbleSituation164

Post doctoral researcher specializing in astrochemistry and astronomical data science with particular expertise in machine learning, scalable analysis of large observation datasets, and development of computational pipelines for radio and astrochemical dataset generation.

Education

University of Virginia, Ph.D. in Chemistry Aug 2019 – July 2025

Thesis: *From Code to Cloud: Machine Learning Approaches to Complex Interstellar Chemistry*

Advisors: Eric Herbst & Anthony Remijan

University of California, Davis, BS in Chemistry Sept 2013 – Dec 2018

Degree emphasis in Environmental Chemistry

Research Experience

Postdoctoral Research Fellow, Univ. of British Columbia – Vancouver, BC Sept 2025 – Current

- Development of machine learning pipeline for the prediction of astrochemically relevant reaction rate constants using established astrochemical networks
- Construction of robust cheminformatics-based molecular and reactions database for fine-tuning of pretrained foundational chemical models of interstellar chemical systems
- Ongoing work includes end-to-end reaction-level embedding, Bayesian optimization and active learning, and integration of observational constraints for astrochemical modeling

Grote Reber Doctoral Research Fellow, National Radio Astronomy Observatory – Charlottesville, VA Aug 2023 – Sept 2025

- Developed scalable computational workflow to benchmark dimensionality reduction algorithms to improve data compression of high-dimensional radio astronomical image datasets
- Evaluated scalability and interpretability of low-dimensional representations for large-scale interferometric datasets
- Assessed algorithmic performance in capturing critical features of physical structures in low-dimensional representations of observational data

Graduate Student Researcher, Univ. of Virginia – Charlottesville, VA Aug 2019 – Aug 2023

- Applied machine learning pipeline to predict chemical abundances in the Orion-KL nebula as a function of spatial environment to enable data-driven guidance for astrochemical studies
- Assessed supervised learning approaches (decision trees, regression models) for predictive modeling of astrochemical inventories

GOTHAM Large Program, Collaborator – Green Bank Observatory, WV Aug 2021 – Sept 2025

- Contributed machine learning-based models for molecular abundance predictions for the detection of targeted anion species in TMC-1 using the Green Bank Telescope
- Supported development of data-driven frameworks for identifying and prioritizing candidate molecular species in observational surveys

Graduate Student Researcher - Pate Research Group Aug 2019 – Aug 2023
Univ. of Virginia – Charlottesville, VA

- Analyzed chiral and deuterated species using rotational spectroscopy to determine purity and absolute configuration
- Performed validation trials of chiral tagging method for use as a commercialized analytical tool for complex sample analysis

- Implemented reproducible analysis workflows for spectroscopic data processing and interpretation of novel pharmaceutically-relevant deuterated molecules in coordination with synthetic researchers at UVA and Marquette University
- Developed theoretical and laboratory spectroscopic methodologies, including chirped-pulse and 3-wave mixing instrumentation

Grants, Funding, & Awards

Grote Reber Doctoral Fellowship \$35,000/year (2 years) National Radio Astronomy Observatory	2023
Graduate Research Assistantship \$11,000 National Radio Astronomy Observatory	2023
International Travel Support Award €350 IAU Astrochemistry conference series	2022
GSAS Teaching Relief Fellowship \$11,000 Graduate School of Arts & Sciences, Univ. of Virginia	2022

Teaching Experience

Head Teaching Assistant , Univ. of Virginia – Charlottesville, VA	Jan – May 2022
• Served as the primary graduate teaching assistant for large undergraduate quantum mechanics lecture course • Lead and prepared discussion sections that provided a collaborative environment to review lecture material and problem-solving exercises • Provided feedback to students through open discussion, graded assignments, and office hours • Additional responsibilities included assignment grading, gradebook management, and classroom technical support for hybrid-style course	
Laboratory Teaching Assistant , Univ. of Virginia – Charlottesville VA	August 2019 – May 2021
• Served as a graduate teaching assistant for the undergraduate physical chemistry laboratory course • Lead laboratory experiments and worked alongside students to complete capstone projects • Presented completed analysis and projects at an international spectroscopic conference on behalf of students • Additional responsibilities included assignment and report grading, instrumentation upkeep, sample preparation, and laboratory management and safety	

Additional Experience & Service

Peer Review Reviewer for journals including <i>The Astrophysical Journal</i> and <i>ACS Earth and Space Chemistry</i>	
Graduate Council Representative Graduate School of Arts & Sciences, Univ. of Virginia <i>Finance committee</i>	Aug 2021 - May 2024
Dept. of Chemistry Graduate Student Council Officer Director of Graduate Studies Ambassador, Aug 2022 - May 2023 Head Organizer of the Graduate Invited Speaker Lecture, Aug 2022 - May 2023 Graduate Student Mentor, Aug 2020 - May 2022	
Dept. of Chemistry Graduate Outreach Chemical education and demonstration outreach – L.E.A.D. at UVA	Aug 2019 - May 2023

Programming & Computational Proficiencies

Programming & Software Engineering

- Python, LaTeX
- Git, GitHub
- CI/CD

Machine Learning & Data Science

- PyTorch, scikit-learn
- Dimensionality reduction
- Image segmentation
- Bayesian optimization

Scientific Computing

- HPC environments, SLURM, cluster computing
- Astropy, CASA, CARTA
- Gaussian16, Psi4

Presentations

Cosmic Horizons Conference Charlottesville, VA, USA	July 14, 2026
The 75th International Symposium on Molecular Spectroscopy Univ. of Illinois – Urbana-Champaign, IL, USA	June 25, 2026
American Chemical Society Astrocheminar Series Virtual webinar	<i>Invited talk</i> Nov 19, 2025
American Chemical Society Spring 2025 Meeting San Diego, CA, USA	<i>Invited talk</i> Mar 23-27, 2025
Advanced Data Products Workshop European Southern Observatory – Garching, Germany	Dec 10-12, 2024
Green Bank Observatory Science Community Webinar Green Bank Observatory – virtual webinar	July 19, 2023
Kavli - IAU Astrochemistry Symposium Traverse City, MI, USA	July 10-14, 2023
From Clouds to Planets II: The Astronomical Link Harnack Hause – Berlin, Germany	Oct 3-7, 2022
The 75th International Symposium on Molecular Spectroscopy Univ. of Illinois – Urbana-Champaign, IL, USA	June 20-24, 2022
The International Symposium on Molecular Spectroscopy (Virtual) Univ. of Illinois – Urbana-Champaign, IL, USA	June 21-25, 2021

Publications List

- Benchmarking Dimensionality Reduction Methods for High-Dimensional ALMA Image Cubes using Machine Learning** 2026
HN Scolati, RA Loomis, AJ Remijan, KLK Lee *The Astrophysical Journal*
- The Missing Link of Sulfur Chemistry in TMC-1: The Detection of c-C₃H₂S from the GOTHAM Survey** 2025
AJ Remijan, PB Changala, C Xue, EQH Yuan, M Duffy, HN Scolati, CN Shingledecker, TH Speak, IR Cooke, RA Loomis, AM Burkhardt, ZPT Fried, G Wenzel, A Lipnicky, MC McCarthy, BA McGuire *The Astrophysical Journal*
- Exploring Effects of Modified Machine Learning Pipelines of Astrochemical Inventories** 2025
H Toru Shay, HN Scolati, G Wenzel, KLK Lee, AN Marimuthu, BA McGuire *The Astrophysical Journal*
- Explaining the Chemical Inventory of Orion KL with Machine Learning** 2023
HN Scolati, AJ Remijan, E Herbst, BA McGuire, KLK Lee *The Astrophysical Journal*
- Astronomical Detection of the Interstellar Anion C₁₀H⁻ toward TMC-1** 2023
AJ Remijan, HN Scolati, AM Burkhardt, PB Changala, SB Charnley, IR Cooke, MA Cordiner, H Gupta, E Herbst, KLK Lee, RA Loomis, CN Shingledecker, MA Siebert, C Xue, MC McCarthy, BA McGuire *The Astrophysical Journal Letters*
- Absolute Configuration Assignment of Chiral by Virtue Molecules by Deuterium Substitution using Chiral Tag Rotational Resonance Spectroscopy** 2023
ZP Vang, RE Somstrom, HN Scolati, JR Clark, BH Pate *Chirality*
- Rapid Enantiomeric Excess Measurements of Enantioisotopomers by Molecular Rotational Resonance Spectroscopy** 2023
RE Somstrom, ZP Vang, HN Scolati, JL Neill, BH Pate, JR Clark *Organic Process Research & Development*
- Searching for Propionamide (C₃H₇NO) toward SagB2 at Centimeter Wavelengths** 2022
C Schuessler, AJ Remijan, C Xue, JT Carder, HN Scolati, BA McGuire *The Astrophysical Journal*
- Enantioselective Synthesis of Enantioisotopomers with Chiral Analysis by Chiral Tag Rotational Spectroscopy** 2021
MD Mills, RE Somstrom, ZP Vang, JL Neill, HN Scolati, CT West, BH Pate, JR Clark *Angewandte Chemie*